

PROJECT 21

ITL Screen Printer — Voltage-to-Frequency Control Circuit

Effective Engineering (Client: Industrial Technology Institute)
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PROJECT SUMMARY

Designed and tested a custom voltage-to-frequency conversion circuit for the ITL screen printer machine. The circuit was verified using a Haiwell PLC test program before being installed in the live machine and commissioned.

OBJECTIVES

- Design a voltage-to-frequency converter circuit for the screen printer control system.
- Verify circuit function with a PLC test program before live installation.
- Support machine commissioning at ITL.

HOW IT WORKS

The screen printer required a specific high-frequency signal for one of its control functions. A custom V-to-F circuit was designed and bench-tested using a Haiwell PLC (T16S2T) with push buttons and a simple ladder test program to confirm correct output before being installed in the machine.

KEY COMPONENTS

Component	Specification / Function
V-to-F Converter Circuit	[TO BE FILLED] — IC and passive components
Haiwell PLC T16S2T	Test platform for pre-installation verification
Push Button + E-Stop	Test inputs for circuit verification

TECHNICAL SPECIFICATIONS

Parameter	Value	Notes
[TO BE FILLED]	Output frequency range	—
[TO BE FILLED]	Input voltage	—
Haiwell PLC	Test platform	T16S2T

MY CONTRIBUTION

- Designed the voltage-to-frequency converter circuit.
- Wrote a PLC ladder test program and verified circuit output.
- Confirmed circuit performance before live installation.
- Supported integration and commissioning in the ITL screen printer.

CHALLENGES & SOLUTIONS

[TO BE FILLED] — any circuit or commissioning challenges.

RESULTS

Circuit designed, bench-tested, and confirmed working before installation. Successfully integrated into the ITL screen printer control system.

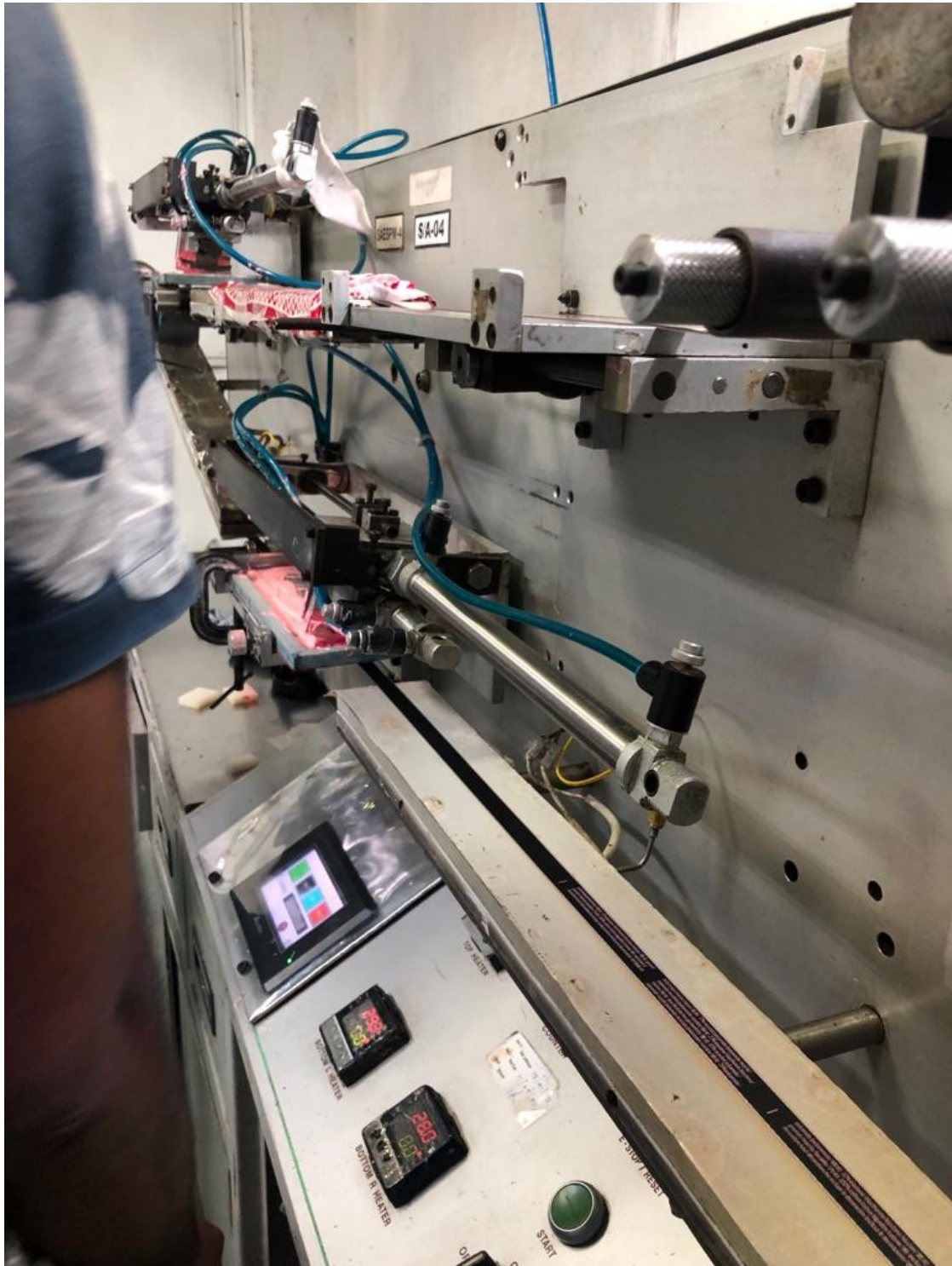
TOOLS & SOFTWARE USED

Tool / Software	Used For
Haiwell Happy PLC Software	Ladder logic test program
Multimeter	Circuit verification
Soldering Iron	Circuit assembly

IMAGES & TECHNICAL DRAWINGS



ITL screen printer machine at site



Screen printer machine — detail view